

Module 4 Summary

Response to feedback

If you are resubmitting, include a statement outlining the changes you have made to your submission. This section can be short but should be precise. It is a good idea to quote the feedback you are responding to.

If this is your first submission, include a statement about what part of the lesson review you would most like to receive feedback (and why). Your tutor will take this into consideration when reviewing your work, although they may choose to give you feedback on a different thing if they think it's more appropriate.

Module Learning Objectives

I certify that I achieved the following learning objectives for the module (these objectives can be found in the introduction of the module):

Calculate determinants using the cofactor expansion

Use properties of determinants (Theorem 3.1.2) to calculate determinants

Calculate matrix inverses using $\text{adj } A$ and $\det(A)$

Identify the inverse of an orthogonal matrix

Perform LU-factorisation on a matrix and use this to solve systems of linear equations

Key Definitions and Theorems

The Cofactor Expansion

Textbook Ref	Theorem/Definition statement	Learning Notes
Determinant Calculation for a 3×3 matrix	For a 3×3 matrix, A – the <u>determinant</u> is given by $\det A = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$ $= aei + bfg + cdh - ceg - afh - bdi$ $= a(ei - fh) - b(di - fg) + c(dh - eg)$ $= a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$	

Definition 3.1 Cofactors of a matrix	Assume that determinants of $(n-1) \times (n-1)$ matrices have been defined. Given the $n \times n$ matrix A, let A_{ij} denote the $(n-1) \times (n-1)$ matrix obtained from A by deleting row i and column j. Then the (i, j)-<u>cofactor</u> $c_{ij}(A)$ is the scalar defined by $c_{ij}(A) = (-1)^{i+j} \det(A_{ij})$ Here $(-1)^{i+j}$ is called the <u>sign</u> of the (i, j)-position.	
Definition 3.2 Cofactor expansion of a matrix	Assume that determinants of $(n-1) \times (n-1)$ matrices have been defined. If $A = [a_{ij}]$ is $n \times n$ define, $\det A = a_{11}c_{11}(A) + a_{12}c_{12}(A) + \cdots + a_{1n}c_{1n}(A)$ This is called the <u>cofactor expansion</u> of $\det A$ along row 1.	
Theorem 3.1.2 Determinant properties	Let A denote an $n \times n$ matrix. 1. If A has a row or column of zeros, $\det A = 0$. 2. If two distinct rows (or columns) of A are interchanged, the determinant of the resulting matrix is $-\det A$. 3. If a row (or column) of A is multiplied by a constant u, the determinant of the resulting matrix is $u(\det A)$. 4. If two distinct rows (or columns) of A are identical, $\det A = 0$. 5. If a multiple of one row of A is added to a different row (or if a multiple of a column is added to a different column), the determinant of the resulting matrix is $\det A$.	

Determinants and matrix inverses

Textbook Ref	Theorem/Definition statement	Notes
Theorem 3.2.1 Product theorem	If A and B are $n \times n$ matrices, then $\det(AB) = \det A \det B.$	
Theorem 3.2.2 Det relationship to inverse	An $n \times n$ matrix A is invertible if and only if $\det A \neq 0$. When it is invertible, we have, $\det(A^{-1}) = \frac{1}{\det A}$	
Theorem 3.2.3 Det of transpose	If A is any square matrix, $\det(A^T) = \det A$	
Theorem 3.2.4 Adjugate formula	If A is any square matrix, then $A(\text{adj } A) = (\det A) I = (\text{adj } A) A$ In particular, if $\det A \neq 0$, the inverse of A is given by $A^{-1} = \frac{1}{\det A} \text{adj } A$	

LU Factorisation

Textbook Ref	Theorem/Definition statement	Notes
Theorem 2.7.1 LU Theorem	If A can be lower reduced to a row-echelon matrix U, then $A=LU$ where L is lower triangular and invertible and U is upper triangular and row-echelon.	
LU Algorithm	Let A be an $m \times n$ matrix of rank r, and suppose that A can be lower reduced to a row-echelon matrix U . Then $A = LU$ where the lower triangular, invertible matrix L is constructed as follows: 1. If $A=0$, take $L=I_m$ and $U = 0$. 2. If $A \neq 0$, write $A_1=A$ and let c_1 be the leading column of A_1. Use c_1 to create the first leading 1 and create zeros below it (using lower reduction). When this is completed, let A_2 denote the matrix consisting of rows 2 to m of the matrix just created. 3. If $A_2 \neq 0$, let c_2 be the leading column of A_2 and repeat Step 2 on A_2 to create A_3. 4. Continue in this way until U is reached, where all rows below the last leading 1 consist of zeros. This will happen after r steps. 5. Create L by placing $c_1, c_2, \dots, c_r$ at the bottom of the first r columns of I_m.	

Summarising your understanding:

- Add notes to the definitions and theorems tables that reflect your understanding of the definition/theorem, how it relates to the learning objectives and things to keep in mind when applying – these notes can be somewhat informal but should be understandable to your OnTrack tutor.

Write a mathematical summary for the module that includes the questions/tasks in the dot points below.

- Summarise Cramer's rule and how it works. Include reference any of the relevant definitions and theorems.
- Include a worked example of applying some of the rules in Theorem 3.1.2 to simplify the calculation of a larger determinant.
- Write a statement on LU factorisation and when it can be applied.

Reflecting on the content:

- Reflect on what you found interesting/difficult in this module, how it relates to previous content covered in this and other units, and which parts helped you to meet each of the learning objectives. This should be written from your personal perspective and not just read as a general summary of the topic.